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Modeling the oxidation-induced fragmentation of soot aggregates in laminar flames

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Abstract

While the formation and growth of soot particles has received much attention, the subsequent destruction of the particles is less well understood. Soot particles are destroyed though two parallel processes: oxidation and fragmentation. Oxidation is the removal of mass from particles due to chemical reactions with molecular oxygen and hydroxyl radicals. Fragmentation is the break-up of large aggregates into smaller aggregates. Here, a new model for fragmentation inspired by previous experimental investigations is proposed and formulated within the Hybrid Method of Moments (HMOM). With the formulation, the rate of particle loss due to oxidation is closed, resolving a long-standing problem with the Method of Moments. Less important, secondary unclosed terms are introduced, and models for these terms are proposed. The oxidation and fragmentation models are validated using a set of laminar premixed methane flames and then applied to a series of laminar counterflow diffusion acetylene flames. The role of fragmentation is distinctly different in the two flame types. In the premixed flames, fragmentation only occurs in lean flames with a high oxygen concentration. In the diffusion flames, fragmentation is virtually absent, for soot passes through an OH radical layer and is completely oxidized before reaching high oxygen concentrations.
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1. Introduction

Throughout the years, the understanding of soot formation and growth processes has led to models with increasing predictive capability in both simple laboratory-scale configurations as well as engineering devices. However, much less is known about the destruction of soot particles. The models describing destruction are rather empirical, and their predictive capability is not at the same level as models for formation and growth. Arguably, in practical combustion systems in which the emission of soot is undesirable, predicting particle destruction is as important as predicting particle formation and growth.

Two processes lead to the destruction of soot particles: oxidation and fragmentation. Oxidation is the process by which oxidizing species from the gas phase (predominantly hydroxyl radicals and molecular oxygen) react with the surface of soot particles, removing carbon mass. Fragmentation is the process by which large aggregates break apart into small pieces, presumably after the structure has been weakened by oxidation.

Kinetic rates for the oxidation of soot have been studied widely. The most commonly used rate

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for oxidation by OH, the predominant oxidizing species in most configurations, is that of Neoh et al. [1], though other rates have been proposed [2,3]. For oxidation by O_2, the most commonly used rate is that of Nagle and Strickland-Constable [4]. However, the rate is somewhat complicated, and the simpler formulation of Kazakov et al. [5] is commonly used. Other O_2 oxidation rates have been proposed which can vary substantially from the former two rates (see, e.g., [6]). However, in many flames, including those considered in this work, oxidation by O_2 is a relatively small portion of the total oxidation, and the soot volume fraction is insensitive to the kinetic rate.

Fragmentation has received much less attention in the literature than oxidation. The fragmentation of soot was first experimentally observed by Neoh et al. [7,8]. Soot was generated in a rich methane burner and subsequently burned in a premixed flat flame of the diluted combustion products. More recently, a similar configuration has been studied by Echavarria et al. [9]. In both experiments, under lean conditions, the number density was observed to increase. Based on this observation, Neoh et al. hypothesized that molecular oxygen, since it reacts more slowly than hydroxyl radicals, is able to penetrate the particle structure before reacting. Once the structure has been sufficiently weakened, the particle breaks apart. Oxidation-induced fragmentation was later also observed by Garo et al. [10] in a laminar methane co-flow diffusion flame. The idea of internal burning inspires the model for fragmentation in this work.

Computationally, the effect of fragmentation on the soot population was investigated by Harris and Maricq [11]. In their work, it was determined that the inclusion of a model for fragmentation resulted in better agreement with experimental measurements of the size distribution of soot particles in the exhaust of a diesel engine. However, the form of the fragmentation rate was borrowed from other disciplines in which fragmentation is induced by fluid shear which should be unimportant at the scales of soot particles and which was absent in the laminar premixed flames studied by Neoh et al.

In this work, a model for soot fragmentation is developed based on the idea of internal burning by molecular oxygen. A brief overview of oxidation is also given including the kinetic rates used in this study. Both of the models are formulated within the Hybrid Method of Moments (HMOM), so requisite details regarding this method are first presented. With the formulation, the particle loss rate due to oxidation is closed, resolving a long-standing issue with the Method of Moments as discussed below. Other unclosed terms are introduced, which are not as important as the former terms, and closures are proposed. To validate the models, a set of methane premixed flat flames is considered. These are the same set of flames in which Neoh et al. first experimentally observed the fragmentation of soot. Finally, the model is applied to a set of acetylene counterflow diffusion flames to examine the role of fragmentation in a very different combustion environment.

2. Hybrid Method of Moments

The soot population is described by the Number Density Function (NDF) N. In order to account for fractal aggregates, two quantities are required to describe a particle, the volume (V) and the surface area (S) [12]. With this description, the population can be a mixture of single spheres and aggregates composed of individual primary particles. The NDF is therefore defined on a two-dimensional internal state space. Subsequently, N_i will refer to the number density of particles of a given size, that is, a specific (V_i, S_i) pair.

The evolution of the NDF is governed by the Population Balance Equation (PBE). Source terms in the PBE include contributions from particle nucleation from Polycyclic Aromatic Hydrocarbon (PAH) dimers, particle–particle coagulation, condensation of PAH dimers on particles, and surface growth by acetylene in addition to oxidation and fragmentation which are discussed in subsequent sections. Details regarding the modeling of these terms can be found in Mueller et al. [13] and the references therein.

Moment methods solve equations for the moments of the NDF rather than the NDF itself. While complete information about the NDF is lost, these methods are computationally much less expensive than methods which attempt to reconstruct the NDF. In addition, the lower order moments (means and variances) are the most practically relevant quantities. The moments of the NDF are given by

$$M_{x,y} = \sum_i V_i^x S_i^y N_i, \tag{1}$$

where summation over i implies summation over the two-dimensional VS state space, including all possible discrete combinations of V and S, and the pair x, y is the order of the moment in volume and surface area, respectively. When the above summation is applied to the PBE, evolution equations for the moments are obtained. The source terms of the moment equations are unclosed, requiring moments which are not solved for.

The Hybrid Method of Moments (HMOM) [13] is a combination of two earlier moment methods: the Method of Moments with Interpolative Closure (MOMIC) [12,14–17] and the Direct Quadrature Method of Moments (DQMOM) [18,19]. In MOMIC, unknown moments are determined from polynomial interpolation of known moments. The major benefit of MOMIC is its

simplicity and numerical robustness. However, the complete method has not been extended to a joint multi-variate description of soot particles, and Mueller et al. [12] have shown that the extension of just a portion of the method to a fully multi-variate description is inadequate for reproducing the moments of strongly bimodal distributions. In DQMOM, the NDF, for the sake of computing unknown moments, is approximated by a series of delta functions. While DQMOM was shown to be quite accurate especially for multi-variate descriptions [19], the algorithm requires the frequent inversion of an ill-conditioned matrix.

In HMOM, the two methods are combined with each method representing the contribution from one of the two modes of the typically bimodal soot NDF. The contribution from the larger particles, the second mode, is represented with polynomial interpolation, and the contribution from the smaller particles, the first mode, is represented with a delta function. The resulting method is able to account for the bimodality without incurring the additional computational cost of the frequent matrix inversion. The closure of the moments is then given by

$$M_{x,y} = N_0 V_0^x S_0^x + \exp\left(\sum_{r=0}^{R}\sum_{k=0}^{r} a_{r,k} x^k y^{r-k}\right). \quad (2)$$

In the above expression, the first term corresponds to a delta function fixed in state space at a location corresponding to the size of the smallest soot particles (V_0, S_0). In addition to a set of moments, an equation is also solved for the weight of the delta function N_0. In the equation for N_0, the source term is given by

$$\dot{N}_0 = \lim_{\alpha,\beta\to\infty} \frac{\dot{M}_{-\alpha,-\beta}}{V_0^{-\alpha} S_0^{-\beta}}. \quad (3)$$

Finally, the interpolating coefficients $a_{r,k}$ in the second term of Eq. (2) are determined from the moments which are solved for. Complete details regarding HMOM, including the derivation of the source term for N_0, can be found in Mueller et al. [13].

3. Oxidation

In this paper, two oxidizing species will be considered: hydroxyl radicals and molecular oxygen. The kinetic rates for these processes are specified in two different ways. For hydroxyl radicals, Neoh et al. [1,7] found that the rate of oxidation is well approximated by the collision rate between soot and hydroxyl radicals multiplied by a constant efficiency. For molecular oxygen, the reaction is modeled as an integrated part of the surface chemistry. In addition to oxidation, particles grow by the addition of mass through the H-Abstraction, C_2H_2-Addition (HACA) mechanism [20]. In that mechanism, reactions take place at hydrogenated sites (or radical sites) on the surface. Blanquart and Pitsch [19,21] have developed a model in which the number of hydrogenated sites is an additional independent descriptor of a particle, the Volume-Surface-Hydrogen (*VSH*) model. However, it is not known how the density of hydrogenated sites changes during oxidation. Therefore, in the present work, the density of these sites is assumed to be proportional to the surface area with the number of sites per unit area given by $\chi(= 1.7 \times 10^{19}\ \mathrm{m}^{-2})$. The rate for oxidation by molecular oxygen is taken from Kazakov et al. [5]. The kinetic rate parameters for oxidation are given in Table 1. As discussed previously, while numerous kinetic rates have been proposed in the literature, only the most commonly used rates are considered in this work.

With the kinetic rates specified, the source term for the NDF is specified as the difference between a production rate and a consumptions rate:

$$\dot{N}_i^{\mathrm{ox}} = k_{\mathrm{ox}}(S_{i+1}N_{i+1} - S_i N_i), \quad (4)$$

where the parameter k_{ox} is a function of the kinetic parameters of the oxidation rates, species concentrations, temperature, and the number of hydrogenated sites per surface area χ. In the production rate, S_{i+1} and N_{i+1} are the surface area and number density of particles that, when oxidized, become size (V_i, S_i). Taking the moment of the source term and taking advantage of the fact that the change in volume δV and surface area δS are small compared to the size of a particle, the moment source term is then given by

$$\dot{M}_{x,y}^{\mathrm{ox}} = -k_{\mathrm{ox}} \frac{\delta V}{V_0} V_0^x S_0^{y+1} N_0 - \sum_{i=1} k_{\mathrm{ox}} \left(x\frac{\delta V}{V_i} + y\frac{\delta S}{S_i}\right) V_i^x S_i^{y+1} N_i. \quad (5)$$

In the above expression, the first term corresponds to the removal of the smaller particles represented by the delta function, and the second term corresponds to the reduction in size of the larger

Table 1
Kinetic rate parameters for oxidation reactions. Parameters refer to the modified Arrhenius form ($k = A\exp(-E/RT)$). All units are in cm, mol, s, and kJ. Soot $- H$ refers to hydrogenated sites, and Soot $- *$ refers to radical sites.

Reaction	A	E	Ref.
Soot $- * + O_2 \rightarrow$ Soot $- * + 2CO$	2.20×10^{12}	31.38	[5]
Soot $- H + OH \rightarrow$ Soot $- H + CO$	Efficiency: 0.13		[1]

particles, where summation from $i = 1$ implies only the contribution of the second mode is considered. The removal of particles from the population is a long-standing issue with the implementation of oxidation with the Method of Moments. With other methods, the term is either neglected [16] or crudely modeled [22]. However, with HMOM, the removal of particles from the population is straightforward. Following the work of Blanquart and Pitsch [19], during oxidation, the number of primary particles in an aggregate is assumed to be constant. The change in surface area is then given by

$$\frac{\delta S}{S} = \frac{2}{3}\frac{\delta V}{V}. \tag{6}$$

Taking the limit in Eq. (3), the source term for the weight of the delta function is given by

$$\dot{N}_0^{\mathrm{ox}} = -k_{\mathrm{ox}}\frac{\delta V}{V_0}S_0N_0 + k_{\mathrm{ox}}(S_0 + \delta S)N_1. \tag{7}$$

The first term is the destruction of the small particles (same as the first term in Eq. (5)), and the second term corresponds to the intermodal transfer of particles from the second mode to the first as the larger particles are oxidized. To close this latter term, it is assumed that the number of particles lost from the second mode is proportional to the mass lost from the second mode:

$$N_0^{\mathrm{inter,ox}} = C_{\mathrm{inter,\ ox}}k_{\mathrm{ox}}\delta V M_{-1,1}^L, \tag{8}$$

where the superscript L refers to the contribution to the moment from the second mode. To specify the parameter $C_{\mathrm{inter,ox}}$, it is assumed that the transfer rate is large if an average second mode particle is comparable in size to a first mode particle and small if an average second mode particle is much larger than a first mode particle. Therefore, the parameter is assumed to be the volume ratio between the two modes (V_0/V_L). It was found that the results presented in the next section, which were validated with Monte Carlo simulations (not shown), are not sensitive to the specification of the parameter.

4. Fragmentation

The fragmentation source term for the NDF is once again expressed as the difference between a production and a consumption rate:

$$\dot{N}_i^{\mathrm{frag}} = \sum_j \gamma_{i,j}k_{f\,j}N_j - k_{f\,i}N_i, \tag{9}$$

where k_f is the fragmentation rate. In the production term, $\gamma_{i,j}$ is the fragmentation pattern which specifies the number of particles of size (V_i, S_i) formed when a particle of size (V_j, S_j) fragments. Both of these parameters must be specified. Taking the moment of the source, the source term for the moment equations is generally expressed as

$$\dot{M}_{x,y}^{\mathrm{frag}} = \sum_i \left(\sum_j \gamma_{i,j}k_{f\,j}N_j - k_{f\,i}N_i\right)V_i^xS_i^y. \tag{10}$$

The effect of the fragmentation pattern on the overall size distribution was investigated computationally by Harris and Maricq [11] using a sectional method. Two patterns were considered: one in which fragmentation results in two particles of equal size and one in which fragmentation results in one particle with twice the mass of the other. They determined that the resulting size distribution was rather insensitive to the fragmentation pattern. Therefore, the simpler of the two approaches will be taken here, and it is assumed that a particle fragments into two particles of equal volume and equal surface area. The fragmentation source term then becomes

$$\dot{M}_{x,y}^{\mathrm{frag}} = \left(2^{1-x-y} - 1\right)\sum_i k_{f\,i}V_i^xS_i^yN_i. \tag{11}$$

In the above expression, the factor multiplying the summation is a consequence of the fragmentation pattern.

Since soot aggregates break apart as a result of internal burning by molecular oxygen, the fragmentation rate should be related to the O_2 oxidation rate. To determine the fragmentation rate, consider the following *notional* argument. An aggregate has grown, through condensation, coagulation, and surface growth, such that the volume between the primary particles has been filled with mass. Molecular oxygen oxidizes the aggregate until there is a point of weakness between two of the primary particles, and the aggregate breaks apart into two pieces. It is further assumed that the point of weakness is nearly a point contact, so the surface area does not appreciably change as a direct result of fragmentation, as indicated in Eq. (11) above. The fragmentation rate is then given by

$$k_f = \frac{k_{\mathrm{O_2}}\chi\pi\left(n_p - 1\right)d_p^2}{\delta V}, \tag{12}$$

where $k_{\mathrm{O_2}}$ is the O_2 oxidation rate coefficient, δV is the volume that must be removed to cause fragmentation, and d_p and n_p are the diameter and number of primary particles comprising an aggregate, respectively. From the *notional* argument, if it is assumed that oxidation is more or less uniform, the change in volume is given by

$$\delta V = \frac{\pi}{12}\left(n_p - 1\right)d_p^3. \tag{13}$$

While constant factors appear in the rate as a result of the *notional* argument, it is the scaling that is the more important result. The complete fragmentation source term is then given by

$$\dot{M}_{x,y}^{\text{frag}} = \left(2^{1-x-y} - 1\right) \sum_{i=1} 12 \frac{k_{O_2}\chi}{d_{p_i}} V_i^x S_i^y N_i. \quad (14)$$

Taking the limit of Eq. (3), the source term for the weight of the delta function is given by

$$\dot{N}_0^{\text{frag}} = 24 \frac{k_{O_2}\chi}{d_{p_1}} N_1. \quad (15)$$

The only term in this expression is the transfer of particles from the second mode to the first mode as the larger particles break apart. It is assumed that this rate is proportional to the overall fragmentation rate:

$$\dot{N}_0^{\text{inter,frag}} = C_{\text{inter,frag}} 12 k_{O_2}\chi M_{-1,1}^L. \quad (16)$$

As for oxidation, the parameter $C_{\text{inter,frag}}$ is taken to be the volume ratio between the two modes (V_0/V_L).

5. Validation and application

Two cases are considered for the validation of the model and examination of the role of fragmentation during the oxidation of soot. The first case is a set of methane premixed flames measured by Neoh et al. [7,8]. The purpose of these first comparisons is to validate the model in the same experiment in which the phenomenon was first observed. The second case is a set of acetylene counterflow diffusion flames studied by Mauss et al. [22]. In this flame, a partially premixed fuel stream opposes a pure oxygen stream. Soot is formed in the premixed flame region and is transported toward the oxygen rich region, where it is oxidized in a diffusion flame. Specific details about the configurations are given below.

HMOM with the models for oxidation and fragmentation is implemented as part of the FlameMaster code [23]. All simulations are performed with seven soot scalars (six moments plus the weight of the delta function). This number of moments is the optimal balance between accuracy and computational cost. The chemical mechanism used has been created to accurately predict soot precursors and has been extensively validated in a variety of configurations for a broad range of fuels [24,25].

5.1. Methane premixed flat flames

Neoh et al. [7,8] studied soot oxidation and fragmentation in a series of laminar premixed methane flames. The experimental system consisted of two burners: a primary fuel-rich premixed burner and a secondary premixed flat flame burner for oxidizing the soot generated in the primary burner. The gases exiting the primary burner were quenched and mixed with additional oxygen and nitrogen to vary the global equivalence ratio. In this work, simulation results will be presented for two global equivalence ratios, one lean and one rich, which show significant differences in soot evolution. The flow rates, equivalence ratios, and temperatures for the two flames that will be considered here are given in Table 2. The other three flames measured, one slightly leaner than the lean case here ($\phi = 0.85$) and the other two between the two cases here ($\phi = 1.05, 1.10$), have been computed and will be discussed below, but the results will not be presented due to space limitations.

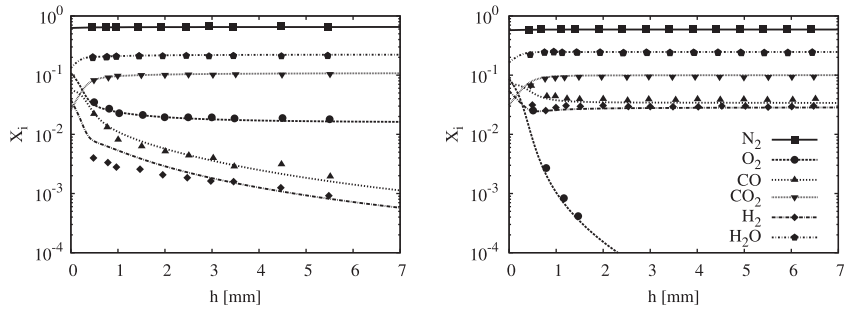
In order to generate boundary conditions for the secondary burner, the primary burner is modeled as a burner-stabilized premixed flat flame. The experimental mass flux is imposed in the simulation, but the temperature is computed, including the effects of gas-phase radiation. The eight most prolific components of the products (H_2, H_2O, CO, CO_2, C_2H_2, CH_4, C_2H_4, and C_6H_6) are diluted with the appropriate amounts of oxygen and nitrogen and used as boundary conditions for the second burner. All chemical reactions are neglected during the quenching and mixing processes. The eight components make up to 99.9% of the primary products by volume. For the secondary burner, both the experimental mass flux and temperature profiles are imposed. In order to minimize the uncertainty in the soot boundary conditions for the secondary burner, a Monte Carlo simulation is performed for the primary burner, although the differences between the Monte Carlo and the HMOM results are relatively minor. The moments of the resulting Monte Carlo distribution, after dilution, are then used as boundary conditions for the HMOM simulations in FlameMaster.

The mole fractions of the major species for the two flames are shown in Fig. 1. Overall, the agreement between simulation and experiment is quite good and remains within experimental uncertainties. The major difference between the two flames is the presence of the species leading to fragmentation. In the rich flame, molecular oxygen disappears rapidly, and fragmentation induced by O_2 oxidation should not be important.

The soot volume fraction is shown in Fig. 2a. The soot volume fraction compares very well to the experiments, with the oxidation rate in the lean flame being only slightly overpredicted. This discrepancy is likely due to errors in the OH

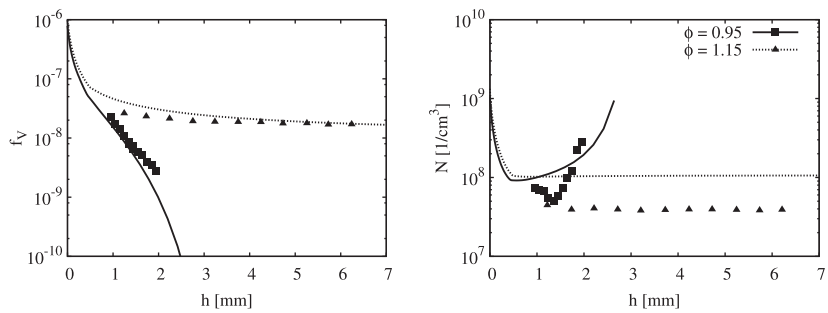
Table 2
Parameters for the two-burner system of Neoh et al. [7,8]. For the secondary burners, the equivalence ratio is the global equivalence ratio. T_{10} refers to the temperature 10 mm above the burner.

Burner	Flow rate (L/min)			ϕ	T_{10} (K)
	CH_4	O_2	N_2		
Primary	2.126	2.024		2.1	1601
Secondary		2.452	12.40	0.95	1746
		1.673	9.138	1.15	1561



(a) $\phi = 0.95$ (b) $\phi = 1.15$

Fig. 1. Mole fractions of major species in the secondary flames. Lines are simulations, and symbols are experiments.



(a) Volume Fraction (b) Number Density

Fig. 2. Mean soot quantities in the secondary flames. The number density is for *qualitative* comparison only. Lines are simulations, and symbols are experiments of Neoh et al. [7,8].

concentration as oxidation by OH accounts for nearly 99% of the total oxidation rate. In addition, discrepancy arises due to uncertainty in the kinetic rate for OH oxidation, with the oxidation rate being linear in the kinetic rate. Since O_2 oxidation plays such a minor role, the volume fraction results are not sensitive to the kinetic rate for O_2 oxidation. Similar agreement is obtained for the additional three flames measured. More importantly, the decrease in the oxidation rate (due to a reduction in OH concentration) is well reproduced. Additionally, it was found that fragmentation does not significantly affect the volume fraction.

The number density from the simulations and a qualitative measure of the number density from the experiments are shown in Fig. 2b. In the experiments, a number density was computed by presuming a size distribution of spheres, so direct quantitative comparisons between the two quantities are not important. While the two quantities are related [26], the qualitative comparison is of more interest here. In the lean flame, in which molecular oxygen is present, the number density is observed to increase due to fragmentation. The fragmentation rate appears to be slightly underpredicted, and this can be attributed to uncertainty in the kinetic rate for O_2 oxidation, with the fragmentation being linear in this kinetic rate, as well as the specification of the constant in Eq. (14). Conversely, in the rich flame, in which there is no molecular oxygen, the number density is constant. In addition to the results shown, the model captures the decrease in fragmentation as the equivalence ratio increases.

5.2. Acetylene counterflow diffusion flame

Mauss et al. [22] studied the effect of strain rate (as well as the carbon to oxygen ratio) on soot formation and oxidation in an acetylene counterflow flame. For the flame considered here, the fuel stream ($h = 0$ mm) is a rich premixture (C/O = 0.85) of acetylene and air opposed by a stream of pure oxygen ($h = 25$ mm). Soot is formed on the rich premixed side and is oxidized in the subsequent diffusion flame of the combustion products. The relevant parameters of the flame are given in Table 3. Two additional flames achieving higher strain rates by increasing the cold gas velocity of each stream by a factor of two and three are also considered. For the purposes of this work, the increase in strain rate is a surrogate for turbulent flames of increasing Reynolds number in which the flame front is more vigorously strained, possibly leading to local extinction.

Table 3
Cold gas velocities v_0 and gas compositions for the baseline acetylene counterflow diffusion flame of Mauss et al. [22].

Stream	v_0 [cm/s]	$Y_{C_2H_2}$	Y_{O_2}	Y_{N_2}
Fuel	4.0	0.1388	0.2006	0.6606
Oxidizer	2.0	0	1	0

Comparisons with measurements of soot volume fraction are shown in Fig. 3a. Overall, the comparisons are good, with too little soot at the lower strain rates. The reduction in volume fraction with increasing strain rate due to an increase in the temperature of the premixed flame (see, for example, Blanquart and Pitsch [21]) is well reproduced. The number density in the three flames is shown in Fig. 3b. Unlike in the premixed methane flame, fragmentation does not seem to be active since the number density is not increasing as the soot is oxidized. Near the end of oxidation in the higher strain rate flames, the number density increases slightly, indicating some fragmentation, as shown in Fig. 3b. However, the effect of fragmentation is not nearly as large as in the premixed flame, with only about a 50% increase in number density near the completion of oxidation in the highest strain rate flame compared to an order of magnitude increase in the premixed flame. In the diffusion flame investigated by Garo et al. [10], the effect of fragmentation was also less pronounced with only a factor of about two to three increase in number density.

The way in which soot is oxidized in the premixed flame is fundamentally different from the diffusion flame. In the premixed flame, soot and molecular oxygen coexist as the two pass through a region rich in hydroxyl radicals. Where soot and oxygen coexist, the soot aggregates fragment. On the other hand, in the diffusion flame, soot and oxygen are on opposites sides of the OH layer. By the time the soot particles are transported through the OH layer, oxidation is nearly complete, and there are no aggregates left to fragment.

To further illustrate this point, the oxidation rates, represented as an inverse time scale, for each of the oxidizing species are shown in Fig. 4 for the highest strain rate flame (the others are similar). The oxidation time scale is defined as

$$\tau_{ox} = M_{10}/\left|\dot{M}_{10}^{\mathrm{ox}}\right| \tag{17}$$

and is normalized by the local strain rate in the flame (the inverse of which is τ_{flow}). Oxidation is active only when the normalized oxidation rate is greater than unity. As already mentioned, soot encounters an OH layer, where it is completely oxidized, before reaching the oxygen rich region. However, as the flame strain rate increases, the OH layer moves closer to the oxygen rich region, and, in the flame with the highest strain rate, some fragmentation is observed as some of the soot (and some oxygen) is able to break through the OH layer before the soot is completely oxidized. In diffusion flames in which OH oxidation is relatively weaker, whether due to more soot or lower OH concentrations as in a turbulent flame at high

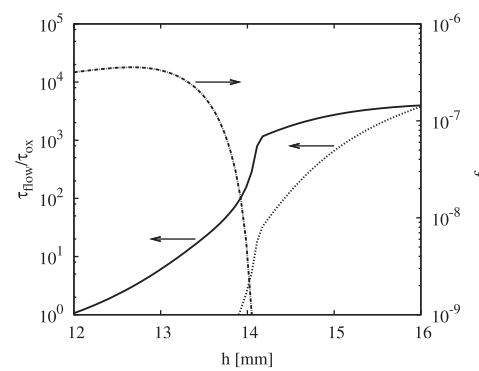
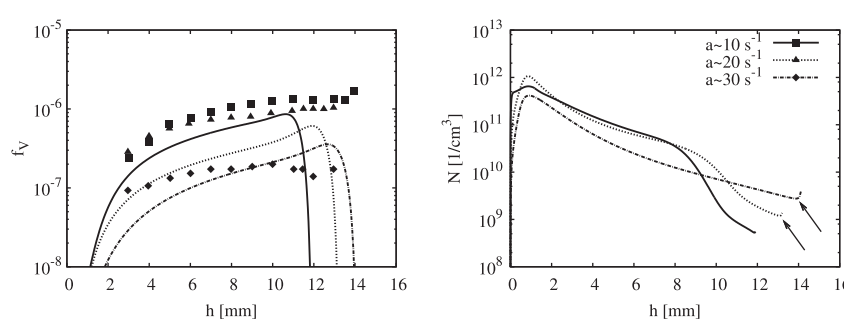


Fig. 4. Oxidation rates and volume fraction in highest strain rate flame ($a \sim 30\,\mathrm{s}^{-1}$). Solid lines are the OH oxidation time scale ratio, dotted lines are the O_2 oxidation time scale ratio, and dashed-dotted lines are the volume fraction.



(a) Volume Fraction (b) Number Density

Fig. 3. Effect of strain rate on soot volume fraction and number density in the partially premixed diffusion flame. Lines are simulations, and symbols are experiments. The arrows indicate fragmentation near the end of the oxidation process at the higher strain rates.

Reynolds number exhibiting local extinction, fragmentation may play a role in the oxidation of soot.

6. Conclusions

In this paper, a model for soot aggregate fragmentation has been developed based on the notion of internal burning by molecular oxygen observed by Neoh et al. An argument was given for the fragmentation rate, which must be proportional to the O_2 oxidation rate to be consistent with the hypothesis of Neoh et al. The oxidation and fragmentation models are formulated within the Hybrid Method of Moments (HMOM). With the formulation, the rate of particle loss due to oxidation is closed, resolving a long-standing issue with the Method of Moments. Less important unclosed terms for the transfer of particles from the second mode to the first mode are introduced and must be modeled. For oxidation, this rate is proportional to the volume loss rate, and, for fragmentation, this rate is assumed to be proportional to the total number loss rate.

To validate the fragmentation model, comparisons were made to the measurements of Neoh et al. In the experiments, soot was generated in a rich premixed methane burner and oxidized in a lean premixed flat flame. The model showed good agreement for the oxidation rate in both a lean and rich flame. In addition, with the model for fragmentation, the number density was shown to increase in the lean flame, consistent with the experimental observations.

As an application of the fragmentation model to analyze the importance of fragmentation in diffusion flames, a series of partially premixed acetylene counterflow diffusion flames was simulated. Unlike the premixed flames, fragmentation was not observed to be as important during the oxidation process. In these flames, soot first passes through an OH layer and is completely oxidized before reaching an oxygen rich environment where the particles can fragment. However, in diffusion flames in which the fuel is more heavily sooting or there are reduced OH concentrations (due to local extinction in a turbulent flame, etc.), fragmentation could play a more significant role in the oxidation of soot.

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